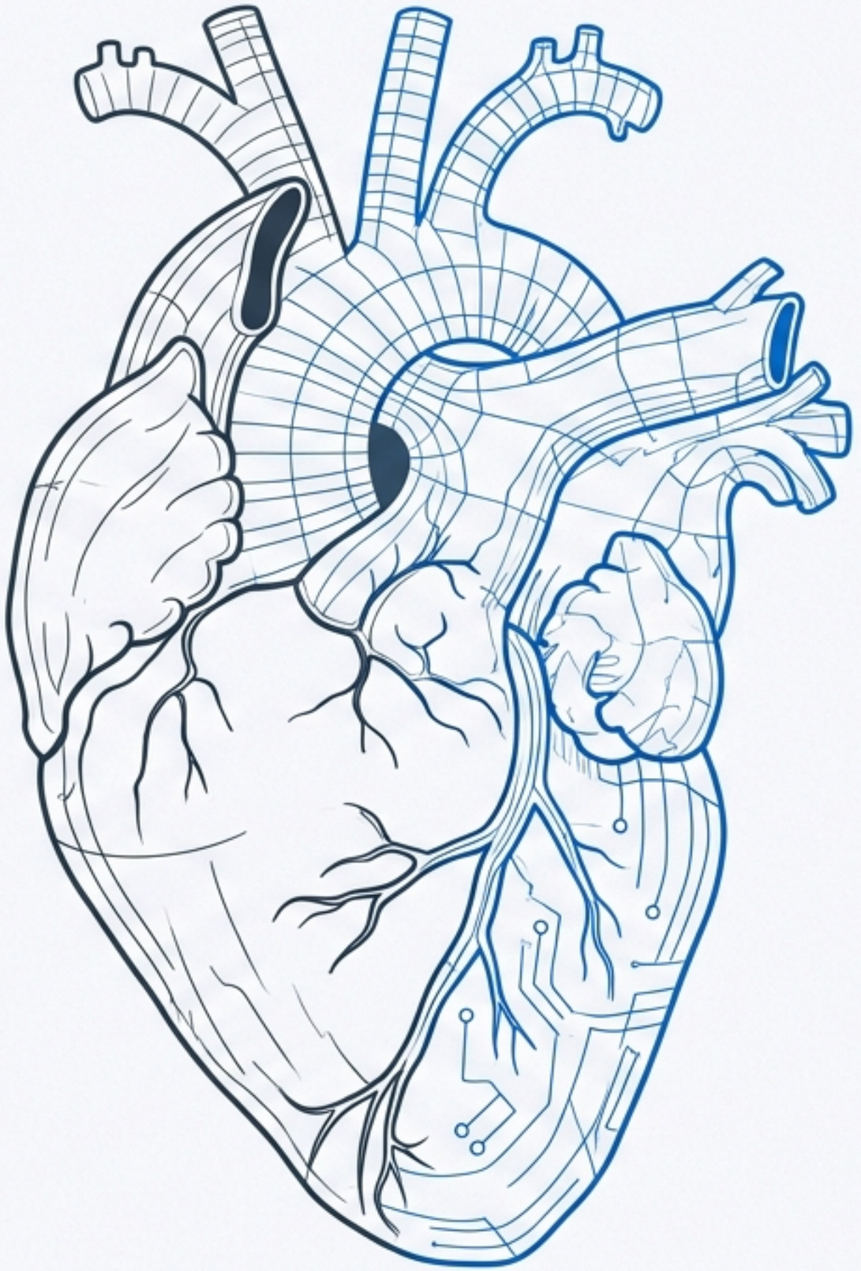
RIET
Rajadhani Institute of Engineering and Technology

Chatbot to Chat with Medical Books using Retrieval-Augmented Generation Model

Motto: MedBuddy: Grounded AI for Evidence-Based Medical Education



Grounded AI | Project Team (RIE23CS)

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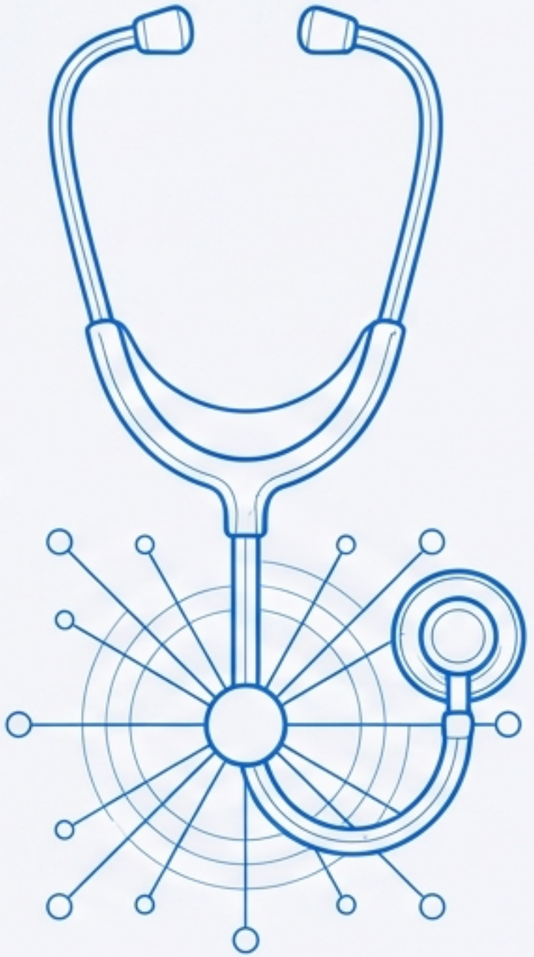
Project Team: & Guide

Project Team (RIE23CS)

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- Akhil Kunduvalappil (RIE23CS011)
- Devika V S (RIE23CS019)
- Pavithra P S (RIE23CS037)
- Prakhar Sharma (RIE23CS039)

Project Guide

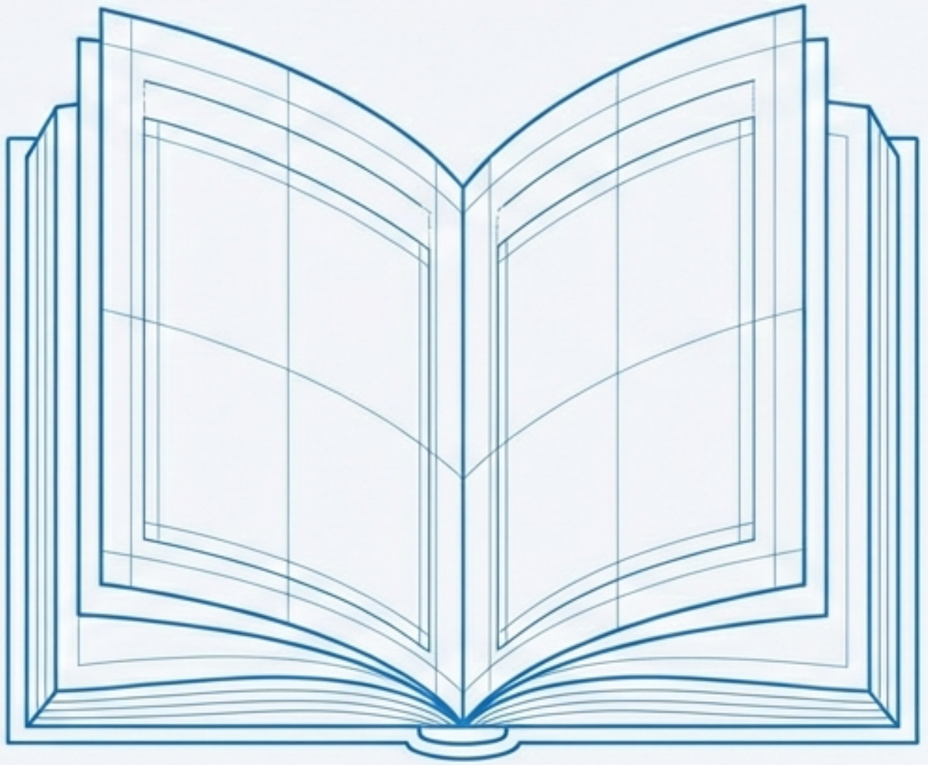
Amrutha S Aravind, Assistant Professor, CSE, RIET



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2. Introduction
3. Literature Survey
4. Problem Identification
5. Proposed System
6. System Software and Requirement
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Abstract

The Core Technology

This project, MedBuddy, is a digital study assistant designed to help medical students. It uses a technology called Retrieval-Augmented Generation (RAG). In simple terms, RAG is a method where the AI searches for the correct answer in a trusted textbook before responding, responding, rather than just guessing. We built this using IBM Watsonx APIs, which provides enterprise-grade, secure Artificial Intelligence tools, replacing less secure public tools.

Technical Implementation

To build this, we used the Python 3.12 programming language managed by a tool called Miniconda, which ensures the software runs smoothly on any computer. We used a 'Lean AI' strategy, meaning we optimized the heavy software libraries (like PyTorch) to run efficiently on a standard laptop CPU without needing expensive supercomputers.

User Experience and Security

The student interacts with a simple website built with Streamlit. Behind the scenes, we use Industry-Standard Secrets Management to keep the API keys (passwords that let the software talk to IBM's servers) safe and hidden. This ensures the project meets the strict security standards required for handling medical data.

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Introduction

The Project Goal

MedBuddy is developed as part of the CSD 334 Mini Project curriculum. The main goal is to solve the “Disk Space Trap” and “Hallucination” problems in AI. “Hallucination” is when an AI makes up false information. MedBuddy prevents this by using trusted medical books as its only source of truth.

Why IBM Watsonx?

Unlike standard chatbots, we switched to IBM Watsonx. This is an "Enterprise-Grade" platform, which means it is designed for businesses and professional use. It allows us to use specific AI models (like the Granite-13b-chat) that, that are optimized to follow instructions carefully and provide concise, accurate answers from the text provided.



The "Open Book" Method

Think of MedBuddy like a student taking an open-book exam.

- Ingestion: It reads the textbook (PDF).
- Indexing: It organizes the information into a mathematical map (Vector Database).
- Retrieval: When asked a question, it finds the exact page.
- Generation: It writes an answer based *only* on that page.

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Literature Survey

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
1	ChEMBL Bot: Chat Bot for ChEMBL Database [2018]	V. Murali et et al.	Created a chat interface to ask questions about chemical compounds and their targets.	Limited only to chemical data queries; it cannot provide guidance from clinical textbooks.
2	Ontology Based Personal Assistive Application [2023]	N. V. Rohil & A. Nair	Used semantic reasoning (understanding meaning) to help elderly people with daily tasks.	Focused on personal daily activity logs rather than academic medical education.
3	Farm-Haystack QA System [2022]	R. S. Nambiar et al.	Integrated Natural Language Processing (NLP) to answer pandemic-related health questions.	Requires heavy 'fine-tuning' (training) for each specific topic and lacks broad clinical utility.

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Literature Survey

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
4	HR based Chatbot using Deep Neural Network [2022]	P. C. Nair et et al.	Used Deep Learning and the Adam optimizer to help with Human Resources communication.	Designed specifically for HR tasks; lacks the specialized medical knowledge required for healthcare.
5	HR based Chatbot using Deep Neural Network [2022]	P. C. Nair et al.	Used Deep Learning and the Adam optimizer to help with Human Resources communication.	Designed specifically for HR tasks; lacks the specialized medical knowledge required for healthcare.
5	AI Chatbot for Nursing Students [2022]	Jeong-Won Han et al.	Developed a chatbot program to improve nursing skills for electronic fetal monitoring.	Restricted to a single nursing skill focus; cannot answer general medical retrieval questions.

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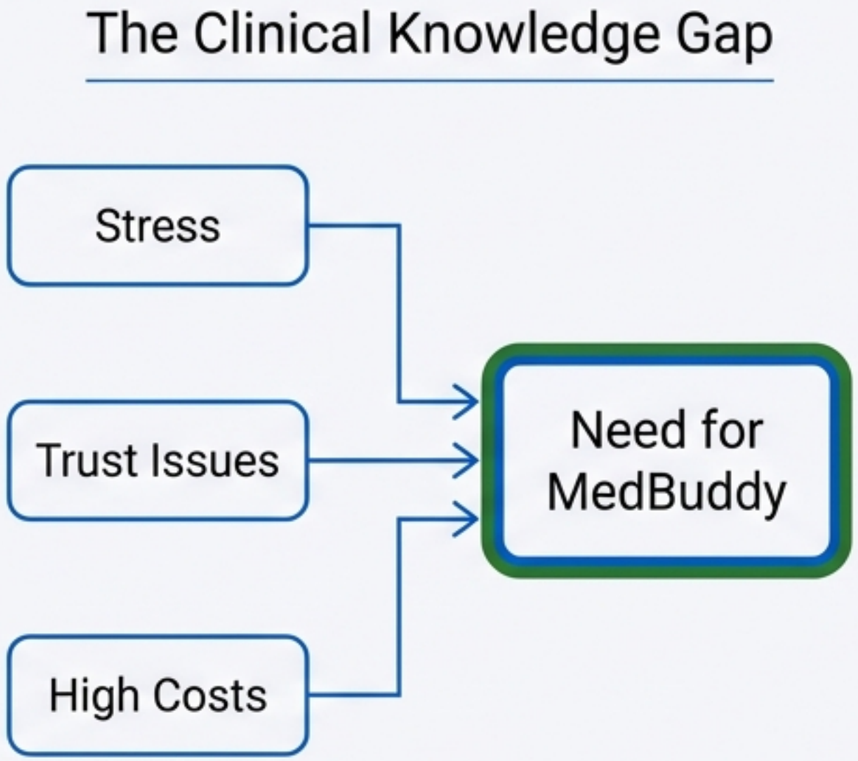
Literature Survey

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
6	The Role of ChatGPT in Medical Education [2023]	Gunther Eysenbach	Investigated the capabilities of large language models like ChatGPT in clinical settings.	Raised significant concerns regarding content authenticity and the risk of 'hallucinations' (fake facts).
7	Exploring AI Chatbots in Public Health [2023]	Francesco Baglivo et al.	A feasibility study on using chatbots for vaccination training.	Requires heavy professional verification to avoid the risk of spreading misinformation.
7	Exploring AI Chatbots in Public Health [2023]	Francesco Baglivo et al.	A feasibility study on using chatbots for vaccination training.	Requires heavy professional verification to avoid the risk of spreading misinformation.

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Problem Identification

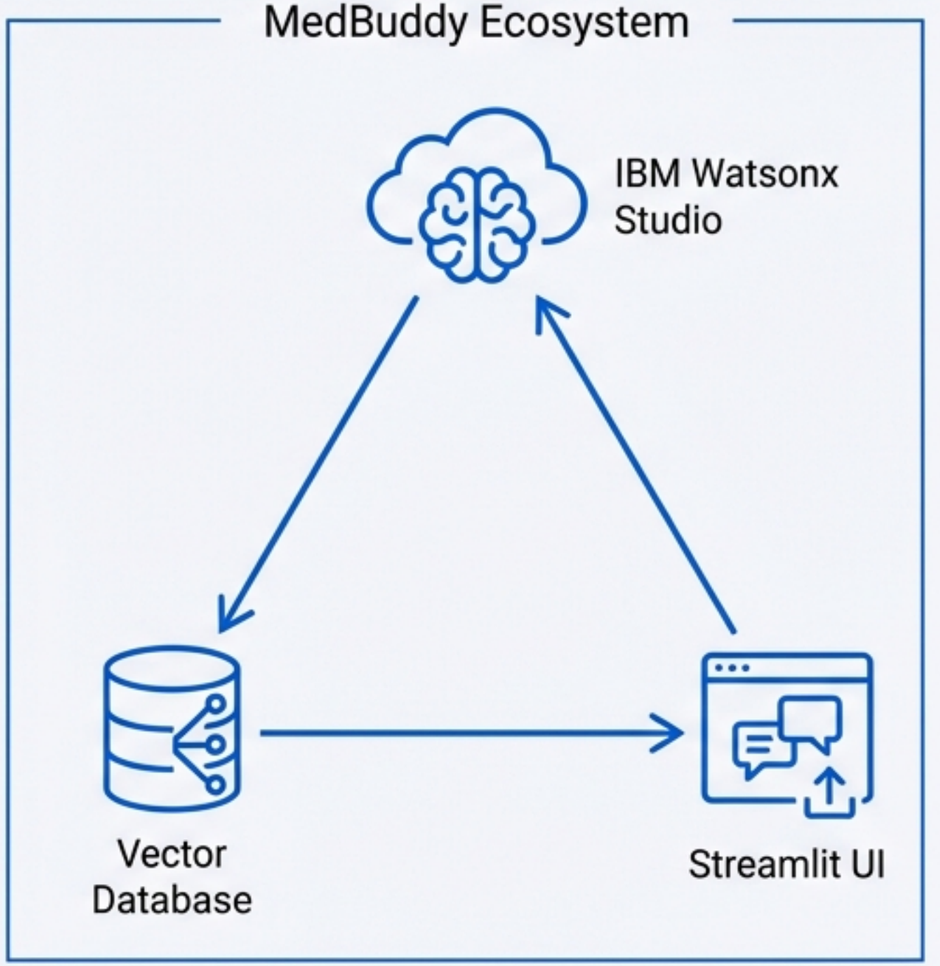
- **Information Overload:** Medical students face massive workloads. Searching through 1,000+ page PDFs manually is slow and stressful.
- **The Trust Gap (Hallucinations):** Generic AI tools often guess when they don’t know the answer. In medicine, a ‘guess’ is a safety risk.
- **Inefficient Search:** Standard ‘Ctrl+F’ keyword searches fail to understand the context of complex clinical questions (e.g., matching symptoms to a diagnosis).
- **Resource Inequality:** High-quality, updated medical textbooks are expensive. This limits access for students in resource-constrained areas.



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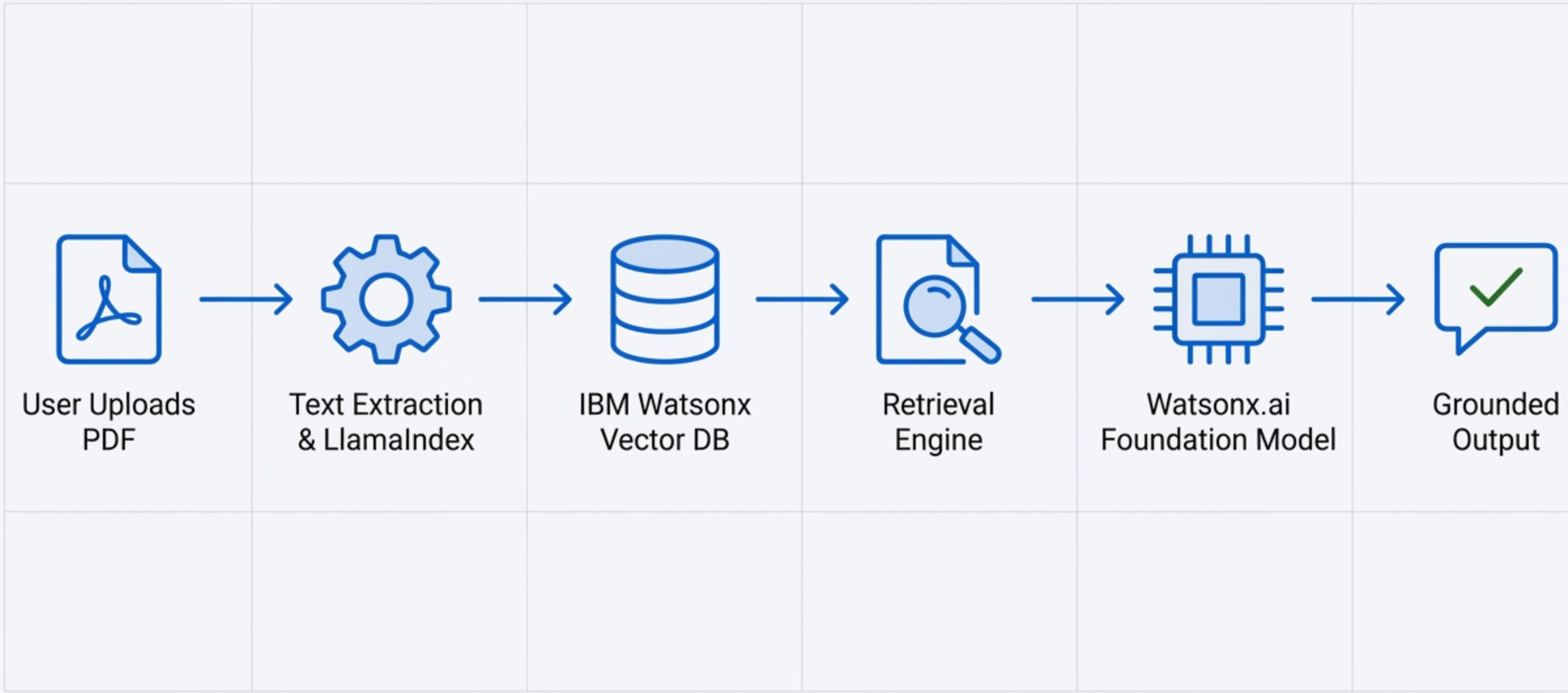
Proposed System

- **IBM Watsonx.ai Studio:** We use this to access the AI "brain." It utilizes AutoAI pipelines, which are automated workflows that ensure the AI's predictions are over 95% confident.
- **Vector Database:** This is a special storage system. It ingests massive datasets (like PubMed or textbooks) and converts the text into mathematical embeddings (lists of numbers that represent meaning) for fast searching.
- **Streamlit UI:** This is the user interface. It provides a clean dashboard where students can upload PDFs and chat without needing to know any code.
- **Secure Environment**: The system runs in a Conda environment (an isolated software box) using Python 3.12 to prevent crashes and ensure stability.



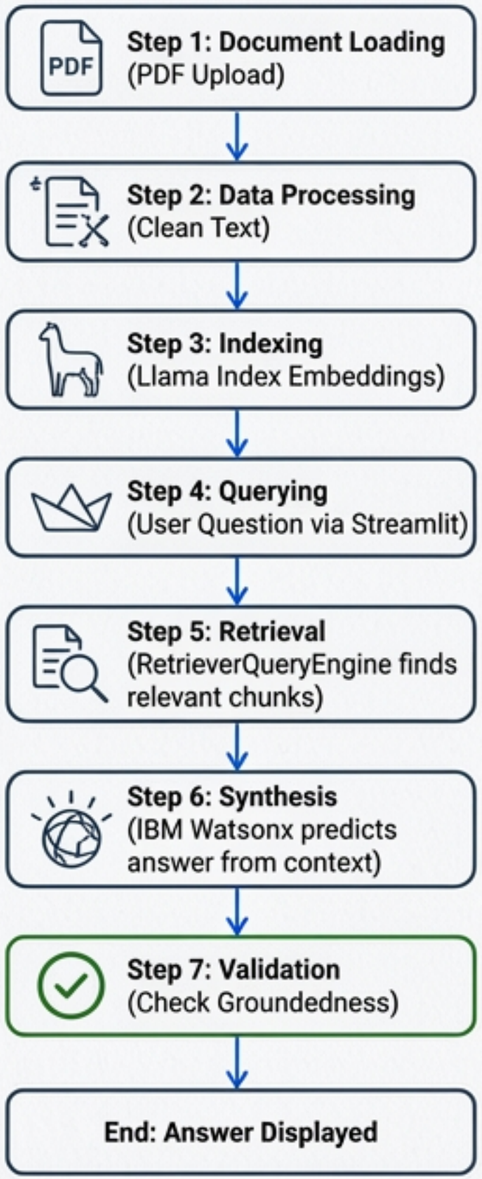
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Architecture



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Algorithm



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Project Plan

Jan 16 - Jan 31, 2026: Data Ingestion & Indexing



- Implement upload pipeline



- Process PDFs to Vector Database

Feb 01 - Feb 15, 2026: Retrieval Engine Development



- Develop fetch logic



- Connect to IBM Watsonx

Feb 16 - Feb 28, 2026: UI Construction & Integration



- Build Streamlit interface



- Backend-Frontend integration

Mar 01 - Mar 15, 2026: Testing & Final Deployment



- Accuracy testing



- Launch on secure cloud

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Future Plan



Multimodal RAG

We plan to upgrade the system to understand not just text, but also medical images like X-rays and MRI scans.



Voice Command Integration

Adding a feature to allow students and doctors to ask questions using their voice, making it easier to use in a busy hospital.



Clinical Co-Pilot

Evolving the tool from a student study buddy into a real-time assistant for doctors, helping them check drug interactions instantly.



Research Acceleration

Using the tool to rapidly summarize massive volumes of new research papers to keep doctors up to date.

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Conclusion



Reduction in Search Time



Groundedness Score (Accuracy)

Summary and Impact

- **Democratization:** MedBuddy makes expensive medical knowledge accessible to everyone, regardless of their location or financial resources.
- **Impact:** MedBuddy successfully bridges the gap between complex AI technology and human needs, creating a less stressful and more effective learning environment for future doctors.

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References

1. V. Murali et al., ‘ChEMBL Bot - A Chat Bot for ChEMBL Database,’’ 2018 14th International Conference on Information Processing (ICINPRO), 2018.
2. N. V. Rohil and A. Nair, ‘Ontology Based Personal Assistive Application,’ 2023 4th IEEE Global Coal Conference for Advancement in Technology (GCAT), 2023.
3. R. S. Nambiar and D. Gupta, ‘Dedicated Farm-Haystack Question Answering System for Pregnant Women and Neonates,’ 2022 12th International Conference on Cloud Computing, Data Science & Engineering, 2022.
4. Peri, S. D. B. et al., ‘Chatbot to chat with medical books using Retrieval-Augmented Generation Model,’ 2024 IEEE ICINPRO, 2024.
5. Baglivo, F. et al., ‘Exploring the Possible Use of AI Chatbots in Public Health Education,’ JMIR Publications, 2023.

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